

Research article

From Greener growth to sufficiency: Modeling alternative global sustainability pathways

Arthur Lauer ^{*} , Luis Llases

Department of Applied Economics, University of Valladolid, Av. Valle Esgueva 6, Valladolid 47011, Spain

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ABSTRACT

This study quantifies two contrasting global sustainability scenarios—FST1: Greener Growth and FST5: Sufficiency Economies—using the MORDRED model. While both pursue high decarbonization without speculative technologies, they embody distinct paradigms: FST1 prioritizes green growth, whereas FST5 centers on sufficiency, redistribution, and well-being within ecological limits. Simulations generate demographic, economic, and biophysical trajectories that complement their qualitative storylines, enabling a plausibility assessment. Results show FST1 achieves rapid technological decarbonization, with a 75% reduction in GHG emissions per unit of output by 2100 and an 83% increase in output over the same period. However, resource and land scarcity constrain growth during the simulation, while existing structural inequalities remain largely unaddressed. As a result, per-capita consumption in the center averages three times that of the periphery, and international climate goals are not achieved, with warming reaching 2.6 °C by 2100. In contrast, FST5 achieves strong convergence in consumption—at €7500 (2019 constant prices)—and mortality rates across regions and classes, stabilizing warming at 1.9 °C by 2100 while reducing material and water extraction and pressures on land systems. At the same time, strong decarbonization increases the renewable sector's share of global resource extraction. Both scenarios appear stable beyond the qualitative storyline's end in 2075, although FST1's greater environmental pressure and inequality pose long-term stability risks. Sensitivity analyses show that plausibility depends on assumptions about climate sensitivity and land and labor productivity growth. This study underscores the value of integrating qualitative narratives with quantitative modeling and highlights that sufficiency-oriented transitions may offer more robust sustainability outcomes.

1. Introduction

Meeting global climate and sustainability objectives requires profound societal transformations and ambitious policy interventions that break with 'business-as-usual' practices, although concrete transition pathways remain contested (Köhler et al., 2019; Scoones et al., 2020; Wiedmann et al., 2020). Sustainability transitions have been linked to Green economies in both the Global North and South (Brown et al., 2014; Hochstetler, 2025; Lederer et al., 2018) as well as to the development of a circular economy (Abunyewah et al., 2023; Fetanat et al., 2021; Flynn et al., 2019), but also to more radical post-growth or degrowth transitions emphasizing the need to overcome the 'growth paradigm', reduce inequalities in all their forms, and achieve a safe and just corridor for all between

^{*} Corresponding author at: University of Valladolid, Avenida Valle Esgueva 6, 47011 Valladolid, Spain.
E-mail address: arthur.lauer22@estudiantes.uva.es (A. Lauer).

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overconsumption and underconsumption (Hickel, 2019; Kallis et al., 2025; Schmelzer, 2017).

Debates about sustainability transitions can be informed and enriched by the development of global environmental scenarios that provide visions for desirable futures but also identify socio-economic and biophysical risks within the scenario space (Juri et al., 2025; Msangi, 2013; Pereira et al., 2021; Wilkinson and Mangalagiu, 2012). Ideally, qualitative and quantitative approaches should complement and inform each other through iterative feedbacks, resulting in scenarios rich in narrative detail and supported by quantitative modeling (Alcamo, 2008). In practice, however, modeling and time constraints often lead to scenario exercises that focus predominantly on either storyline creation or model development.

For instance, many quantitative modeling studies of low-demand or sufficiency transitions lack the narrative context provided by detailed storylines that situate modeled measures within coherent socio-political trajectories (Millward-Hopkins et al., 2020; Wiese, Taillard, et al., 2024, 2024). Moreover, scenario studies with a quantitative component often embody a ‘continuity bias’ (Raskin and Swart, 2020) characterized by the absence of transformative structural changes in the economic, political, and cultural realms that deviate from the status quo (Lauer et al., 2024). For example, strong climate mitigation pathways based on the Low Energy Demand (LED) scenario (Grubler et al., 2018a) or SSP2 (Van Vuuren et al., 2018) do not engage with the institutional and socio-cultural preconditions that would render the modeled policies feasible (cf. Brutschin et al., 2021). Global sustainability scenarios featuring strong and rapid convergence between the Global North and South toward energy levels compatible with meeting basic human needs remain underexplored (Hickel and Slamersak, 2022). Likewise, systematic model-based comparisons of global-scale transformation pathways that envision profound changes in the global institutional environment and world order have yet to be conducted.

Conversely, many qualitative scenarios reflecting more radical approaches to sustainability are characterized by absent or insufficient quantification (Durán et al., 2023; Otero et al., 2020; Sessa and Ricci, 2014), partly due to the limited capacity of existing integrated assessment models (IAMs) to represent radical systemic change. As a result, their concrete implications for material flows, inequality, and environmental outcomes remain invisible.

Further examples for qualitatively rich scenarios lacking a model-based quantitative exploration are the six recently developed ‘fast sustainability transition scenarios’ (FSTs) (Lauer et al., 2025), which narrate alternative sustainability transformation pathways based on competing socio-economic paradigms and approaches to addressing global socio-economic and environmental challenges, including strong convergence and relocalization as well as eco-socialist and green capitalist growth strategies.

This article aims to complement the FST development by quantifying two of the six FSTs (‘FST1 – Greener Growth’ and ‘FST5 – Sufficiency Economies’) using the MORDRED model (Lauer and Llases, 2025). The quantification exercise serves four purposes. First, to enrich the qualitative scenario content with a quantitative dimension that illustrates the rate and scale of the envisioned changes. Second, to assess the extent to which the simulated quantitative dynamics align with the scenario storylines and, where relevant, to uncover unexpected dynamics or contradictions between the qualitative and quantitative versions. Third, to compare projected developments in social, economic, and environmental systems across two scenarios featuring transformative changes grounded in distinct sustainability approaches. Finally, to explore how variations in boundary conditions affect simulation results, thereby narrowing the parameter range within which quantitative dynamics align with qualitative scenario content, enabling a plausibility assessment of both scenarios, i.e. an assessment regarding the internal consistency of the storylines, and their compatibility with current knowledge (Fischer and Dannenberg, 2021).

2. Methods

2.1. Scenario and model description

FST1 and FST5 are two of six global sustainability scenarios describing potential changes in world order, cultural hegemony, (geo) political alliances and global modes of production that align to form global societal transition pathways spanning several decades. We selected the scenarios because they represent fundamentally different approaches to solving the ‘sustainability conundrum’. Although both aim to achieve greater social, environmental, and economic sustainability throughout the 21st century, in cases of conflict between these goals, FST1 prioritizes economic growth over environmental and social objectives, whereas in FST5, well-being—linked to the notion of ‘sufficiency’—is given the highest political priority.

In FST1, by the end of the 2020s, a new historical bloc emerges that successfully advances a global ‘green growth’ agenda, which becomes mainstreamed into economic and social policies with the support of international institutions, civil society, and intellectuals. While the development of green technological innovation is promoted, less attention is paid to social welfare and inequality. Conversely, in FST5, by the late 2020s, a global movement grounded in strong ecological and social values persuades a group of states to advocate for the establishment of a ‘Planetary Confederation’ (a global institution with executive and legislative competences) to fundamentally transform the global economic system and decisively address socio-environmental challenges. This transformation focuses particularly on guaranteeing basic human needs and sufficiency in consumption, enabling voluntary changes in reproductive behavior that gradually reduce the size of the human population to alleviate pressure on planetary systems.

The complete storylines, which narrate power struggles among social actors, shifts in public discourse, and the implementation of diverse sustainability policies, are detailed in Lauer et al. (2025).

The model used to quantify FST1 and FST5 is called MORDRED (Model Of Resource Distribution and Resilient Economic Development). MORDRED is an environmentally and socially extended input-output (IO) model coupled with a demographic model and a climate model, reflecting the growing interest in linking IAM and IO modeling approaches (Lefèvre, 2024; Malik and Schaeffer, 2024). To address the inherently static nature of conventional input-output frameworks, MORDRED treats the input-output coefficient matrix

dynamically, modifying it over time according to scenario assumptions on technological change, efficiency improvements, and energy mix transformations, thereby allowing the economic structure to co-evolve with scenario drivers. Unlike conventional IAMs, which are typically calibrated to reproduce the current economic order and assess policies within that framework, MORDRED features a more generic economic structure that does not presuppose a specific economic system. This enables it to explore a wider range of futures, including scenarios involving radical institutional change, degrowth, or population policies. MORDRED's subcomponents, structure, and data sources are described in detail in the model documentation (Lauer and Llases, 2025).

The model's components represent and project future developments across different systems at the global level. The evolution of the world population results from changes in mortality and fertility rates across three world regions and 20 age cohorts, which are driven by individuals' per capita consumption levels and by public per capita expenditures on health and education. These variables depend on both the region of residence and the individual's social class. At the beginning of each model simulation, the world region 'center'—comprising high-income countries—has the highest per capita health and education expenditures, followed by the 'semi-periphery', consisting of upper-middle-income countries, and the 'periphery', which aggregates all low- and lower-middle-income countries.¹ Within each region, the population fully participating in the global economy is divided into three segments: the richest 20%, the poorest 30%, and the remaining 50%, constituting the middle class. Exogenous scenario assumptions determine how the global distribution of economic consumption goods evolves over the simulation period, enabling the modeling of economic convergence and divergence. Thus, MORDRED can represent different socio-economic groups and inequalities, as well as various forms of distributional justice—an important step toward incorporating justice considerations into IAMs (Low et al., 2025). Importantly, regarding the representation of inequalities, the model focuses on inequalities in terms of consumption goods and services and does not account for inequalities in capital investment and asset accumulation.

Global economic production is represented through Leontief production functions and disaggregated into 25 sectors listed in Supplementary Material 2 (SM2). This sectoral disaggregation allows for the projection of scenario-based developments across different energy sectors, including fossil fuels, biomass, hydropower, solar PV, wind, nuclear, and fossil electricity.

MORDRED's environmental modules project changes in the global climate and land systems and also track the evolution of various forms of air, soil, and water pollution. Additionally, the model converts the economic output of the mining sector into biophysical quantities of extracted materials, including several key minerals for the energy (Owen et al., 2022; Valero et al., 2018).

Importantly, the model incorporates a set of linkages between variables that may constrain economic growth. These constraints operate through several channels. First, labor scarcity arises when the demand for labor hours exceeds the maximum labor supply, which depends on the size of the global working-age population. Second, land scarcity occurs when the demand for specific land types surpasses their available supply. Third, fossil energy scarcity results from the progressive depletion of fossil fuel reserves, ultimately limiting the amount of extractable energy resources per unit of time and increasing extraction costs. Fourth, climate-related damages affect economic performance by reducing the availability and productivity of labor and land, as well as diminishing the capital stock. In addition, production in any sector may decline if the demand for intermediate goods required in the production process cannot be met.

The structure of MORDRED allows scenario developers to selectively activate or deactivate policies, feedback mechanisms, or entire submodules. Depending on the scenario, exogenous parameters can be modified, and new parameters can be added to simulate the effects of different policies - for example, population policies or changes in government expenditure.

Key differences between MORDRED and existing IAMs include the systematic integration of different potential limits to economic growth, the representation of economic inequality, and the link between per capita consumption and demographic variables. Also, while conventional IAMs are 'conservative' regarding social policies, instead focusing on modeling the interplay between technological and economic developments, MORDRED was developed explicitly to explore strong social changes and reflects a more precautionary stance on future technological improvements of productive processes. Thus, MORDRED fulfills most of the minimum requirements for post-growth modeling identified by Van Eynde et al. (2026): (1) it features an economy embedded in the planetary system through the representation of resource use and pollution and feedback mechanisms from the environment onto society; (2) it incorporates limits to technological change and decoupling, disaggregates households and calculates output endogenously; and (3) it represents human time use through modeling working hours, features sectoral disaggregation and can depict specific policies and transformation scenarios. Equally, MORDRED's structural features are aligned with the five core principles of post-growth research, i. e. well-being, sufficiency, reduced inequalities, a repurposing of the economy and north-south convergence (Slamersak et al., 2026). These characteristics render MORDRED particularly suitable to quantify the sufficiency-oriented FST5, given that this scenario postulates strong divergence from historical patterns of growth and inequality, which cannot be meaningfully represented by most existing IAMs.

Using a single model to quantify both scenarios not only simplifies parameterization and simulation but also enhances the comparability of results. Consequently, differences in the development of key variables arise exclusively from differences in scenario parameter assumptions based on their respective storylines, rather than from differences in model structure.

¹ The terms "center", "semiperiphery", and "periphery" are adopted from world-systems theory, which describes the functional roles different regions play within the capitalist world economy (Wallerstein, 2004). In world-system framework, the center concentrates capital-intensive production, appropriates resources from the periphery, and holds a dominant political and cultural position, while the periphery supplies raw materials and cheap labor and remains politically and culturally subordinate. The semiperiphery holds an intermediate position with mixed characteristics, often serving as a buffer between center and periphery. Although these terms should not be reduced to income levels alone, there is significant overlap between a country's position in the world-system and its income level.

2.2. Scenario parametrization

To quantify the FST1 and FST5 storylines, we use the translation protocol outlined in Lauer et al. (2025), which consists of a series of scenario objectives pursued to different degrees in each FST. In every scenario, each objective is assigned a level ranging from 0, indicating that the objective is not pursued, to 4, indicating that it is strongly pursued. Consequently, the main part of the parametrization process consisted in identifying which scenario objectives could be represented in MORDRED and assigning the relevant variables numerical values that reflect the indicated level. In this way, the qualitative differences between FST1 and FST5 were ‘translated’ into quantitative differences by the authors.

When determining the concrete numerical values, we sought to select plausible values, i.e. values that are consistent with the storyline and compatible with current knowledge about possible parameter values, grounded – where possible – in empirical observations. Our underlying approach draws on Herman Daly’s reflection that a ‘biophysical impossibility’ constitutes a harder limit than a ‘political impossibility’ (Daly, 2005): Thus, we assume that under a radically changed institutional constellation, as described in FST5, social changes such as redistribution and behavioral changes could happen in a very short time frame, while even under favorable institutional conditions, technological developments changing the transformation of nature into economic products cannot be scaled up indefinitely and, due to incomplete substitution between different productive factors, are subject to diminishing marginal returns on societal investments.

Nevertheless, a certain degree of subjectivity and uncertainty is an unavoidable part of every scenario parametrization process. Therefore, all parametrized scenario variables for FST1 and FST5 are enumerated in Table 1, allowing researchers to replicate the simulations using other models and/or alternative rates of change for the variables if the values chosen are considered too radical or too conservative.

SM2 includes the numerical values for the main scenario variables, many of which depend on the quantified scenario objectives (Table 1). For the remaining variables, MORDRED’s default calibration was adopted for both scenario simulations. These default values set a common framework for both scenarios and variations in default values do not have significant changes on scenario outcomes.

Following the storylines, we set the beginning of the global societal transition in 2030. The preceding decade (2019–2029) is assumed to represent a ‘business-as-usual’ (BAU) period, marked by limited political effort in the social, environmental, economic, and technological spheres, alongside high consumption demands across all social classes. Therefore, technological and consumption parameters during ‘BAU’ are defined separately and are overwritten by the parameters corresponding to FST1 and FST5 from 2030 onwards. Although the narratives only cover the period until 2075, we simulate them until 2100 to assess their stability beyond the temporal end of the storyline.

Rather than focusing on the projection of future changes within an isolated subsystem of the world, we are interested in the development of the world system as a whole, including interactions and feedback effects between human and natural systems and their impacts on variables featured in FST1 and FST5. Accordingly, all interlinkages and feedback mechanisms between model variables are activated during the simulations.

3. Results

3.1. FST1: Greener growth

The main economic, socio-demographic and biophysical changes during the simulation are shown in Fig. 1, while additional variables are visualized in Section 1.1 of SM1. During the *Greener Growth* scenario, global economic activity continues to expand throughout the 21st century. World output increases by 50% between 2030 and 2100, while the global capital stock nearly doubles over the same period. However, land scarcity constrains economic expansion: compared to a scenario assuming infinite land availability, economic growth in FST1 is considerably lower: by 2075, output in the unlimited-land scenario is almost 1.6 times higher than in FST1 (cf. Section 3 SM1).

The increase in output is linked to growing final demand components. The largest component by far is household (private) consumption, followed by gross fixed capital formation and government (public) consumption. Under *Greener Growth*, both private and public consumption rise significantly during the 21st century, although their growth slows in the second half of the simulation. Thus, between 2060 and 2100, total household consumption increases by only 7 trillion 2019-€, compared to an increase of >23 trillion € between 2020 and 2060, rising from €44.3 trillion to €67.5 trillion. Global public consumption grows by 6 trillion € in the first half of the simulation, but only 1 trillion € in the second, from 19 to 20 trillion €.

Conversely, due to the increasing capital intensity of production and higher depreciation rates from climate-related damages, investment rises almost linearly throughout the entire simulation, reaching 37.6 trillion €/year at the end of the storyline and 45.5 trillion €/year by 2100. The expansion of capital stock and investment is mirrored in industrial output, which grows by 11.2 trillion € between 2029 and 2075, compared to 4.7, 3.3, and 2.2 trillion € in the food, health, and education sectors, respectively (Fig. 1a). Even in relative terms, industry exhibits the highest growth rates, reflecting FST1’s focus on capital accumulation rather than on the development of food, health, and education provisioning systems.

During the transition period (2029–2075), global average per capita consumption rises moderately from 6859 €/year to 8097 €/year, reaching 9152 €/year by 2100. All classes in the center, as well as the richest 20% of the semiperiphery, remain above the global average throughout the simulation, while the poorer 80% of the semiperiphery and periphery never reach it. The top 20% in the periphery, however, grow rapidly and reach the global average around 2040. In 2075, the average regional consumption in the center

Table 1

Quantification of FST1 and FST5 storylines, following the translation protocol outlined in [Lauer et al. \(2025\)](#), which was adapted to take into account specific model characteristics of MORDRED. The variable ‘GHG emission intensity’ was not included in the original translation protocol but was added here since assuming constant GHG emissions would result in a disproportionate increase in non-CO2 GHG such as CH4 and N2O during the simulations compared to CO2. All changes refer to 2100 as target year. All A Matrix related changes, as well as changes in sectoral demand vectors, are linear, whereas changes in the remaining variables are exponential. The numbers and names of the economic sector in MORDRED are contained in SM2.

Scenario objectives	FST1	Quantification	FST5	Quantification
Scale of economic/transformation activities				
Increase scale of consumption	4	Consumption growth rates of 1% (center) to 2.5% (periphery).	0	Consumption growth (1 – 2.5%, depending on the world region) only before the start of the global transition.
Decrease scale of consumption	0	Growth rates only become negative as a consequence of biophysical limits to growth.	3	Consumption per capita decreases and converges to 7500 (2019-)€ between 2030 and 2039.
Reduce size of human population	0	Human population only drops if there is an increase in mortality rates.	1	The endogenous regional birth rates are multiplied by 0.7 (i.e. birth rates decline by 30% across all regions relative to the endogenously calculated birth rates without population policy).
Adapt scale of economic activity to well-being requirements	1	Target consumption levels of poorer classes are increased.	4	Target consumption levels of richer classes are reduced, and of poorer classes increased.
Render scale of economy compatible with energy and material flows of bioregion	No representation of regional energy and material flows			
Increase military & security forces	2	Overall moderate increase, driven by higher demand resulting from general economic growth and a relative decline in the share of security expenditures. The latter is modeled by reallocating 10% of government spending on services (sector 7, including public administration and defense) to education (sector 22) and health (sector 23).	0	Not applicable.
Decrease military & security forces	0	Not applicable (decrease in the share of military and security in total expenses does not amount to an absolute decline).	3	Relative (shift 15% of government expenditures from sector 7 (including public administration and defense) to 22 and 23 (education and health), combined with absolute (due to general degrowth in production) decrease in demand for military and security services.
Average ‘Earth intensity’ of economic activities (depends on sectoral composition and technologies)				
Reduce fossil fuel intensity	4	100% greening of electricity, substitution of 60% of fossil energy with renewable electricity, shift of 50% of remaining fossil energy demands to bioenergy.	4	100% greening of electricity, substitution of 40% of fossil energy with renewable electricity, shift 60% of remaining fossil energy demands to bioenergy.
Reduce GHG emission intensity	2	Emission intensity of CH4, N2O, SF6, HFC, PFC and non-combustion related CO2 reduced by 35%.	2	Emission intensity of CH4, N2O, SF6, HFC, PFC and non-combustion related CO2 reduced by 35%.
Reduce energy intensity	3	Process efficiency and efficiency gains from electrification, depicted through changes in A Matrix.	4	Higher process efficiency due to higher labor intensity (labor dedicated to the optimization of processes) and higher marginal returns on efficiency from electrification due to lower share of electrification.
Reduce land intensity	2	World adopts the productivity levels that the center held in 2019 for the agriculture and forest sectors, contained in sector 1 & 2; land intensity of renewable electricity technologies improves by 30%; land intensity of biomass energy improves by the same rate as the average of agriculture and forest land.	2	World adopts the productivity levels that the center held in 2019 for the agriculture and forest sectors, contained in sector 1 & 2; land intensity of renewable electricity technologies improves by 30%; land intensity of biomass energy improves by the same rate as the average of agriculture and forest land.
Reduce mineral intensity	1	10% reduction in the inputs from sector 4 (mining) in A matrix*.	2	20% reduction in the inputs from sector 4 in A matrix.
Reduce biomass intensity	1	10% reduction in the inputs from sector 2 (biomass) in A matrix.	0	No reduction in biomass intensity is modeled. Intensity is assumed to remain the same as in 2019.
Reduce water intensity	0	No change in water intensity modeled. Intensity is assumed to remain the same as in 2019.	2	20% reduction in water intensity.
‘Circularity’ of economy (duration of materials in economic sphere)				
Increase product use duration	0	No increase in product use duration modeled. Product use duration is assumed to remain the same as in 2019.	4	Multiply input coefficients in A matrix by 0.25 for sectors 5, 11, 7, 21 (industry, manufacturing, services and transport) to represent a 4-fold increase in product duration and use.
Increase reuse of water, biomass and mineral based products	0	No reuse of resources are modeled. Level of reuse is assumed to remain the same as in 2019.	4	Water intensity of industries and households reduced by 50%, inputs from sector 2 and 4 in A matrix reduced by 50%.
Increase recycling of materials (minerals)	1	10% reduction in the inputs from sector 4 in A matrix.	3	25% reduction in the inputs from sector 4 in A matrix.
Labor/Capital-intensity				

(continued on next page)

Table 1 (continued)

Scenario objectives Scale of economic/transformation activities	FST1	Quantification	FST5	Quantification
Increase labor intensity in production process	0	Strong decline in the global average labor intensity to 0.8 times the sectoral labor intensities held by the Center in 2019.	4	Increase of initial global average values by 10%.
Increase capital intensity in production process	4	Capital intensity increases by 30%.	0	Constant capital intensity.
Distribution of consumption goods				
Reduce inequalities in consumption /unequal access to consumption goods	0	No policy-driven changes in consumption inequality.	4	Complete elimination of inter- and intraregional inequality; per capita consumption of the entire world population converges towards €7500 (constant 2019 prices) by 2039.
Aim at need-based allocation of consumption goods	No representation of needs in model.			
Control of productive capacity / the development of productive forces				
Strengthen individual decisions about production	Model does not distinguish between different types of ownership, decision-making and/or ways of production.			
Strengthen collective decisions about production				
Decrease inequality in (private or collective) ownership of means of production				

* Matrix A is the matrix of input–output technical coefficients.

is still three times as high as the global average, converges towards the global level in the semiperiphery, and remains below the global average in the periphery although the gap narrows substantially toward the end of the simulation.

The large population size of the poorest 30% and the middle classes of the semiperiphery and periphery explains the difference between the global average and high consumption levels in the center (Fig. 1c). Rising consumption across all classes reflects the universal aspiration for continuous consumption growth depicted in the FST1 storyline. However, a peak and subsequent decline in consumption can be observed among the center's classes (Fig. 1b), signaling an involuntary break with historical trends and political ambitions. The consumption per capita of richest class of the center does not exceed 50,000 €/ year. After the final year of the storyline, per capita consumption continues to rise in the semiperiphery and periphery but declines gradually in the center, caused by the model's internal allocation rule in case of resource scarcity, which reflects a geopolitical shift not mentioned in the qualitative storyline.

The FST1 storyline also lacks explicit demographic information, which the quantification provides. Model results show that global population peaks around mid-century at 8.8 billion, then begins to decline, reaching 8.7 billion by 2075 - slightly above the 8.3 billion observed at the start of the global transition in 2029. Driven by the development trajectory, global birth rates fall sharply, especially outside the center, leading to a moderate decline in the semiperiphery's population and stabilization in the periphery by 2075. The mortality rate index - calculated for each age group relative to the semiperiphery's base-year value (set to 1) - is projected to decrease for all age cohorts in the semiperiphery and periphery, and to remain constant in the center. Child mortality declines significantly (Fig. 1d), while progress among older populations is more limited. Nevertheless, disparities between center, semiperiphery and periphery persist, reflecting the scenario's reliance on economic development rather than global redistribution, which proves insufficient to eliminate inequalities in life expectancy.

The political efforts to promote the decarbonization of the global economy in FST1 are mirrored in a continuous and fast reduction of total GHG emissions caused by the global sustainability transition (Fig. 1e). Global GHG emissions peak at 54.7 Gt in 2030 and then decline steadily. By 2075, GHG emissions have been reduced by 37% compared to 2030, and by 2100 are >50% below 2019 levels.

The reduction is primarily driven by falling CO₂ emissions. Anthropogenic CH₄ emissions peak with the onset of the transition and decline moderately during the simulation, while N₂O emissions only peak around 2060. However, despite these reductions, the change of global average temperature compared to the pre-industrial period crosses the 2 °C threshold in 2045, reaching 2.4 °C by 2075 and 2.6 °C by 2100.

Notably, emissions decline despite continuous growth in total energy use, reflecting substantial progress in decarbonization driven by a sharp reduction in fossil energy use, including the complete phaseout of fossil-based electricity by 2100, and a strong expansion of electric and non-electric renewables (Fig. 1h), consistent with the FST1 narrative of large-scale investment in green technologies.

However, the transition involves significant increases in mineral demand. Due to the strong increase in renewable electricity, copper extraction rises significantly during the simulation. By 2100, one-third of global copper output serves the renewable energy sector (Fig. 1f). Similar patterns occur for other minerals critical to the energy transition. Bulk materials such as iron and sand also increase steadily, while fossil resources used as industrial feedstocks continue to rise modestly (Fig. 1g). Industrial blue water use more than doubles between 2029 and 2075, and green water use also increases, suggesting that freshwater consumption may exceed ecological thresholds (Rockström et al., 2009). These trends indicate a neglect of non-fossil-fuel-related environmental pressures in the FST1 development paradigm.

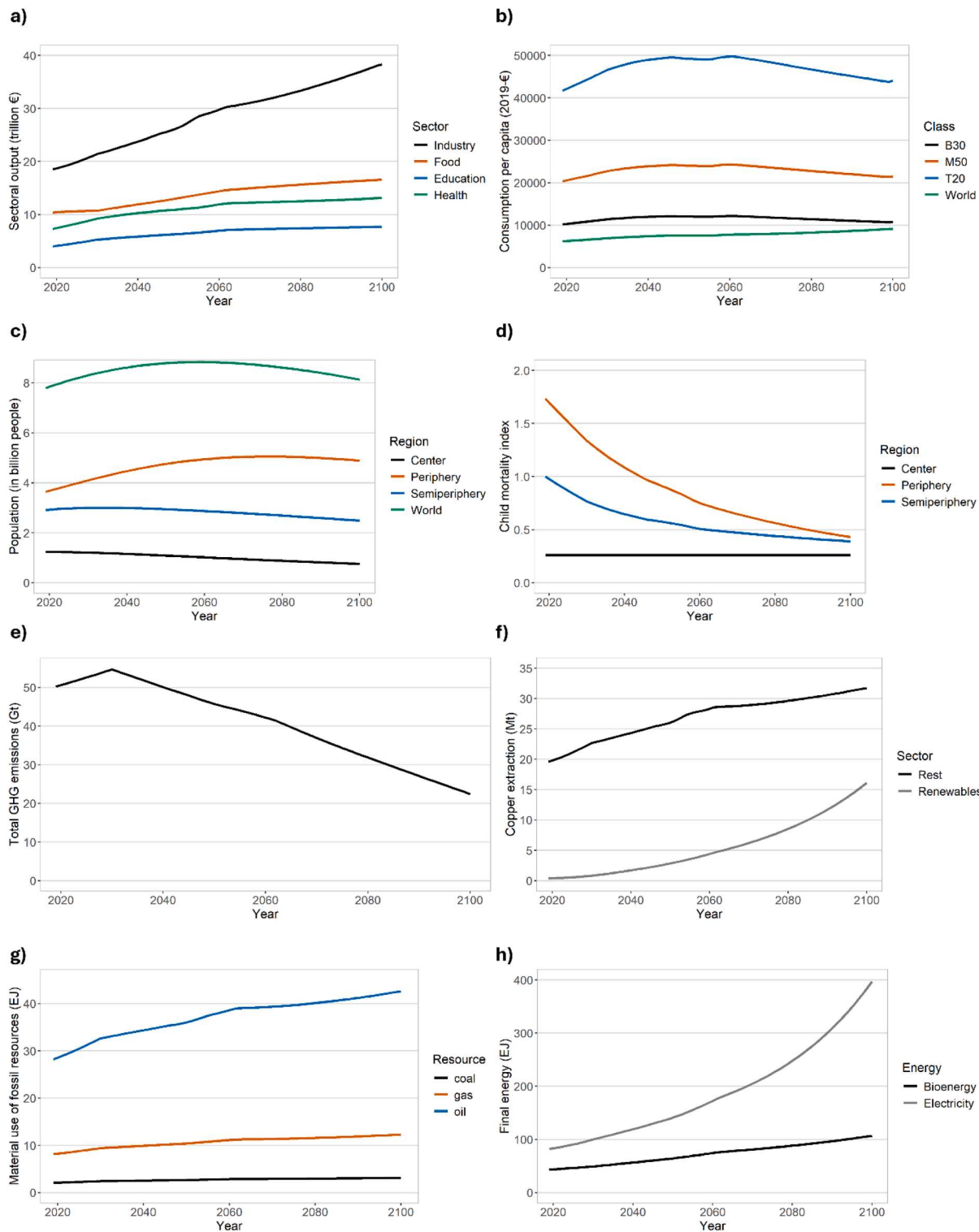


Fig. 1. Evolution of (a) output of the industry, food, education and health sector; (b) average per capita consumption of the world, compared to average consumption of the richest 20%, the poorest 30% and the middle class of the Center; (c) world population, compared to population in the three world regions; (d) child mortality index for the three world regions; (e) annual total GHG emissions without land use change; (f) annual copper extraction for renewable energy technologies and for the rest of economic production; (g) final non-energy use of oil, gas and coal; (h) final (non-electric) energy from biomass and electric energy in the FST1 scenario.

Last, technological changes in FST1 reshape global land systems. Due to the combined land-use pressures from the food and forestry sectors as well as for the expansion of bioenergy and infrastructure, primary forest area declines by 80% over the simulation while land devoted to infrastructure and energy generation expands sharply, occupying 3.4% and 9%, respectively, of global habitable land by 2075.² The slow-down of economic growth is caused by an excessive demand for land to generate renewable energies which stems from the effort to substitute fossil fuels with biomass-based energy, and which cannot be satisfied fast enough.

Hence, compared to the storyline, the quantitative scenario simulation yields some additional insights on 'FST1 – greener growth', notably demographic developments, such as changes in population size and mortality, and projections of biophysical variables enabling a more precise assessment of environmental impacts. Some aspects, however, remain beyond the model's scope, including green trade, speculative technologies (e.g., carbon capture, nuclear fusion), and interregional financial flows.

Overall, the simulation results confirm key shortcomings of the FST1 approach to the sustainability conundrum, as identified in the storyline: higher material extraction rates and increased risks of ecological distribution conflicts (Martinez-Alier, 2003, 2004) due to expanding commodity frontiers, alongside failure to meet the Paris Agreement and rising climate-related damages impacting the food sector. Thus, the title of FST1 is pertinent: Grows becomes greener, especially in terms of decarbonization, yet not green enough to prevent intensifying anthropogenic pressure on planetary systems.

While the storyline does not specify the rate of economic growth, the simulations indicate continued, though slower, expansion of the world economy during the transition, constrained by land scarcity linked to bioenergy expansion. This underscores the growing conflict between accumulation and the ecological imperative in FST1.

On the social side, the simulations confirm FST1's narrative of limited economic convergence. However, the 'burden' of slower economic growth in the simulation is shared between the classes of the semiperiphery and periphery that face a slower decline of poverty, and the classes of the center, that see their consumption levels decline in absolute terms. Thus, in our simulation the weakening of pro-labor policies and unemployment support mentioned in the FST1 storyline mainly affects poorer classes of the center. Since the storyline does not specify how relative losses in growth are distributed across regions, alternative quantifications remain conceivable in which consumption stagnates or declines for poorer classes outside the center, while the center continues to expand.

3.2. FST5: sufficiency economies

In the quantified version of FST5, world output grows to 170 trillion € at the start of the global transition, then declines by 30% and during the next decades increases slowly to reach a second peak at 139 trillion € in 2058. During the second half of the simulation output declines slowly, reaching 109 trillion € by 2075 and 72 trillion € by 2100. Although capital intensity is assumed to remain constant, due to endogenous economic developments that are affected by increasing climate damages and by the growth in relatively capital-intensive sectors, the economic-wide capital-stock-to-output ratio increases moderately during the scenario while the development of total capital stock fluctuates with changing output. As a result of the general decrease in economic activity in FST5, investment approximates public consumption during the transition and by the end of the storyline has even dropped below the public consumption level. Meanwhile, global private consumption reaches a peak in 2058 at 59 trillion € and subsequently declines to 52 trillion by 2075 and 38 trillion € by 2100.

The lower demand for capital goods during the transition drives reductions in the output of the industrial sector. Output in all four sectors peaks between 2055 and 2060 at low levels that do not surpass 12.6 trillion €/year (Fig. 2a). By 2100, with 6.9 trillion €, the output in the food sector is the highest among the four sectors, followed by the health sector. After its peak, the industrial sector's output declines faster compared to the other sectors, and by 2100, is only marginally higher than the education sector's output. These changes indicate the relative importance of food, health and education in FST5 compared to industrial development.

Due to global redistribution policies rapid convergence between classes to the global average consumption can be observed in all world regions. The class that experiences the greatest consumption cuts in absolute terms is the richest class of the center whose consumption drops from almost 45,000 €/year in 2029 to 5243 €/year in 2033 (Fig. 2b). Between 2029 and 2039 the richest class of the semiperiphery as well as the remaining classes of the center lose a significant share of their consumption budget while the poorer classes of the semiperiphery and especially the periphery see their consumption grow at very high rates. As a result of global transition policies, between 2029 and 2055 the per capita consumption of the poorest 30% of the semiperiphery and periphery increases almost 5 and >6.5 times, respectively. The target per capita consumption of 7500 €/year is reached in 2056; by 2059, global per capita consumption has reached 7676 €/year. This value is maintained until the end of the simulation period. Changes in consumption are mainly driven by the input assumptions outlined in Table 1, and slightly affected by land-related limits to sectoral demand.

The consumption reduction of richer classes to levels below the target level, as well as multiple peaks and small instabilities observable in total output, sectoral output and final demand at the end of the 2020s, during the early years of the 2030s and at the end of the 2050s are caused by labor and land scarcity that constrains the desired economic development in FST5. Excessive land and labor demand lead to a lower growth of the poorest class of the periphery, which, in combination with complete and fast consumption convergence, results in global consumption levels lower than the 7500 € target. Since household consumption influences all other relevant economic variables, e.g. public consumption, investment and output, the impacts of scarcity are also observable in the latter. Arguably, a dynamic showing partial convergence during the first three decades of the transition, which would prevent the

² The decline in primary forest does not necessarily imply complete deforestation of primary forest but can also mean that primary forest becomes economically managed and used forest.

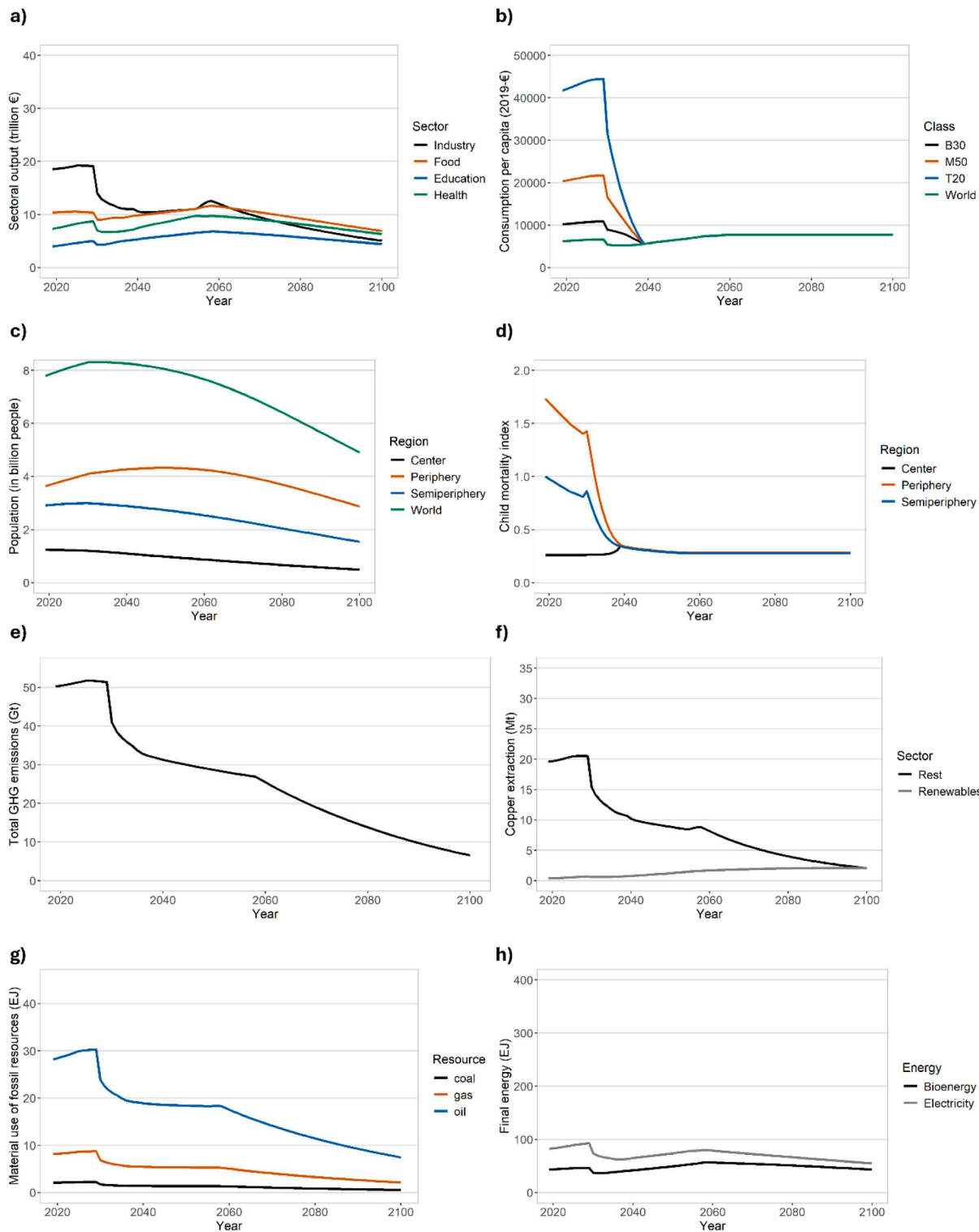


Fig. 2. Evolution (a) output of the industry, food, education and health sector; (b) average per capita consumption of the world, compared to average consumption of the richest 20%, the poorest 30% and the middle class of the Center; (c) world population, compared to population in the three world regions; (d) child mortality index for the three world regions; (e) annual total GHG emissions without land use change; (f) annual copper extraction for renewable energy technologies and for the rest of economic production; (g) final non-energy use of oil, gas and coal; (h) final (non-electric) energy from biomass and electric energy in the FST5 scenario. Additional socio-ecological variables are visualized in [Section 1](#), Supplementary Material 1.

consumption of richer classes from falling below the €7500 target, while allowing poorer classes to gradually approach the target consumption level, can be considered more likely. However, an undesired side effect of this slower convergence would be an increase of overall anthropogenic pressure on planetary systems and a reduction in climate mitigation.

Due to the population policy implemented in FST5, world population already peaks in 2033 and afterwards declines to 6.78 billion people by the end of the storyline. The periphery is the only world region where population continues to grow despite the population policy. However, this trend is reversed in 2049. As a result of these demographic developments, in 2075 the population of the periphery is 5 times as large as the population of the center, and >1.5 times as large as the population in the semiperiphery (Fig. 2c). Combined with the convergence in per capita consumption this implies a much higher weight of the periphery in the 'Planetary Confederation' of FST5. Thus, countries of this world region would likely no longer be 'peripheral' to the global economy but become its central driving force.

Mortality rates in all age groups and regions converge to a global value by 2039, which is significantly lower than initial mortality in the semiperiphery and periphery but slightly higher than initial mortality levels in the center for child mortality (Fig. 2d) and considerably higher than the initial center-values for the mortality of the older population. This points to the importance of an analysis disaggregated not only by class but also by age when studying the effects of economic development on mortality and fertility.

During the first 10 years of the global sustainability transition in FST5 total GHG emissions fall drastically. The reduction slows down between 2040 and 2060 and accelerates again during the second half of the simulation period (Fig. 2e). This dynamic is driven mostly by global CO₂ emissions and is reinforced by CH₄ and N₂O emissions which start to fall from 2060 onwards, driven especially by a decline in the food sector's output, as well as falling GHG emission intensities.

The statement in the FST5 storyline that the transition to a Planetary Confederation occurs too late to reach the 1.5 °C climate goal is confirmed in the simulation. Nevertheless, it is a remarkable achievement of FST5 to keep global warming below 2 °C during the entire simulation period: The global sustainability transition manages to slow down the increase in global average temperatures, and consequently, between 2029 and 2075 global average temperature only increases by 0.5 °C. By 2100 the world is 1.9 °C warmer than in the pre-industrial (1850) reference period. Thus, FST5 succeeds in keeping global warming within the 2 °C goal, avoiding dangerous climate change. However, even under this ambitious mitigation pathway, there are risks of exceeding multiple climate tipping points (Armstrong McKay et al., 2022).

The transition manages to leave the extent of the world's primary forests unchanged, which is partly explained by the low peaks in the land shares of human-built infrastructure and land used for energy generation (Section 1.2 SM1). Additionally, the growth in industrial blue water consumption slows down during the transition and eventually global blue water extraction declines. By 2100, industrial blue water consumption is projected to be 48% lower than in 2019. The final energy use of fossil resources declines steeply during the entire transition period while renewable electricity grows to reach 20.9 EJ, 21 EJ and 13.1 EJ in 2075 for wind, solar PV and hydropower, respectively. By 2100, fossil-based electricity has disappeared, while the non-electric energetic use of coal, gas and oil has been reduced by 93% compared to the start of the transition. The growth in electric- and non-electric energy does not offset the degrowth of energetic fossil resource use, resulting in a strong net reduction of energy use in FST5.

The growth in renewables compared with the general degrowth in the economy and the assumed improvements in material intensity and material recycling leads to a strong increase in material extraction for the renewable sector and a simultaneous strong decline in material extraction for the rest of the economy. For example, by 2100, half of the total copper production is dedicated to the maintenance of the renewable electricity infrastructure (Fig. 2f).

At the same time, the use of fossil resources as material feedstock decreases, although at a lower rate compared to reductions in fossil fuel combustion (Fig. 2g). By 2075, the world still relies on fossil resource extraction to use the equivalent of 12.7 EJ of oil, 3.7 EJ of gas and 0.9 EJ of coal in industrial processes. The small amount of coal used as industrial feedstock, coupled with the reduction of coal in the electricity sector points to a strong loss of significance of the coal sector in FST5.

Overall, the simulation outcomes fit the FST5 storyline: The egalitarian distribution of wealth is reflected through strong convergence in household consumption while investment in the model exclusively serves to realize private and public consumption and does not lead to increasing inequalities over time through private capital accumulation. The convergence in living standards and mortality rates is strong and fast, while the extraction of both rare and bulk materials is reduced throughout and beyond the scenario, despite the expansion of the renewable sector. The simulation shows a low but permanent reliance on fossil resources for material uses, as well as GHG emissions and annual material extraction rates greater than zero, which points to the absence of an 'ideal' circular economy with closed material circles.

Nevertheless, some additional insights can be gained from the quantification: First, the climate damages represented in the model, which increase with global average temperature, are not significant enough to pose a threat to economic reproduction during and beyond the FST5 scenario storyline, despite the lower labor productivity in this scenario and the negative impact of global warming on labor productivity (Dasgupta et al., 2021).

Second, temporal land and labor scarcity issues can endanger the feasibility of FST5. In other words, higher land and labor productivities, or at least faster growth rates in the first half of the simulation, would likely create a smoother development pathway under FST5. The simulation also indicates that fast growth rates in the poorer classes – such as over 20% in the poorest classes at the global level – might pose feasibility problems.

Additionally, the scenario quantification demonstrates that FST5 only works with significant improvements in land productivity because at low consumption levels, food expenses make up a relatively large share of total expenditure. Thus, an increase in consumption will generate a higher demand for land than a similar increase at a higher consumption level. If land productivity improvements do not match the consumption growth of poorer classes, land conversions and the associated loss of biodiversity and increase in GHG emissions become more probable.

Finally, a dynamic that reduces anthropogenic pressure on ecosystems and sustains high levels of climate mitigation beyond 2075 is the continuous population decline, which is not explicitly considered in the FST5 storyline. This demographic trend, caused primarily by declining birth rates throughout the world, emerges as an important stabilizing factor in the scenario. With the exception of temporary labor shortages in the first half of the simulation, no significant labor scarcity is observed in the second half, despite the aging population. However, workforce utilization remains high throughout the simulation and increases slowly but steadily over the last four decades, suggesting that long-term stability of FST5 will ultimately require either population stabilization or renewed growth in labor productivity.

3.3. Scenario comparison

Comparing the two scenario simulations we find that FST1 is characterized by low growth while output dynamics in FST5 could be described as ‘post-growth and degrowth’ (Fig. 3a). Importantly, FST5 is linked to considerably lower levels of anthropogenic pressure on planetary systems. While FST5 manages to keep global warming below 2 °C by 2100, in FST1 none of the climate objectives of the Paris Agreement are reached (Fig. 3b). Additionally, FST1 has a narrow focus on climate mitigation and ignores other environmental stressors such as water consumption, the material use of fossil resources, material extraction or land use conversion, while in FST5 these variables improve during the scenario. Last, the effect of the presence or absence of limits to growth imposed by the availability of land is much higher in FST1 than in FST5 (Section 3 SM1).

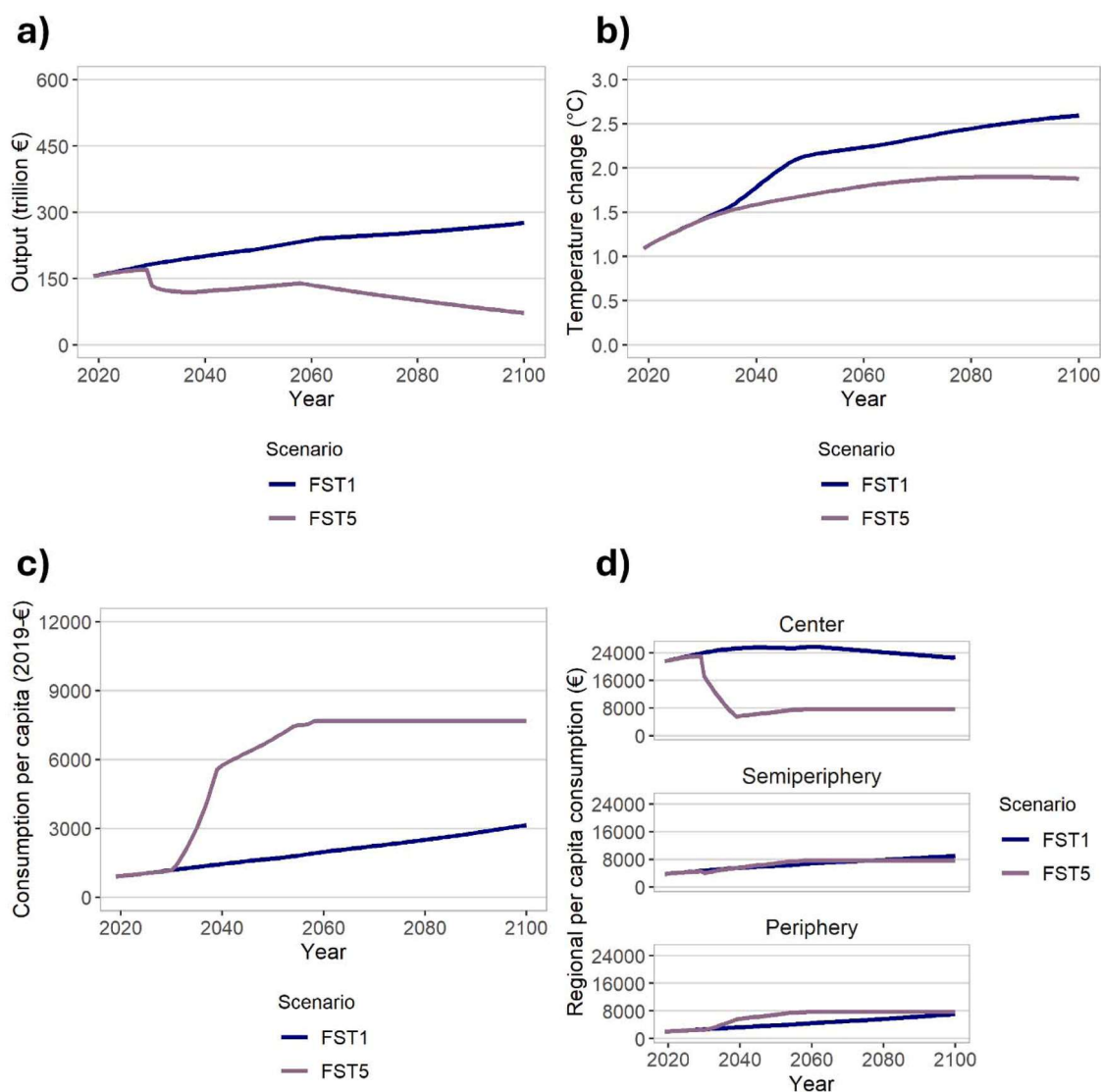


Fig. 3. Evolution of (a) world output; (b) global average temperature change compared to 1850; (c) consumption per capita of the poorest 30% in the Periphery; (d) average per capita consumption per region in the FST1 and FST5 scenarios.

While life expectancy and per capita consumption of the poorest increase in both scenarios, class and territorial inequalities are reduced considerably faster in FST5 (Fig. 3d). Thus, in FST1, the poorest 30% of the periphery live with very low consumption levels throughout and beyond the global transition (Fig. 3c). At the same time, although the richest classes in the center eventually see their consumption levels decline in both scenarios, the loss in consumption is much more pronounced in FST5.

Once the global-level policies are implemented, both scenarios are stable beyond the final year of the storyline (2075), although the greater level of environmental pressure and inequality in FST1 pose mid- and long-term risks for its stability.

Comparing FST1 with the quantified SSP1 scenario (Van Vuuren et al., 2017) we find that projected increases in GDP are comparable between SSP1 and FST1 with no land-use constraints: GDP in SSP1 is projected to increase to about 600 trillion US-\$ by 2100 while the 600 trillion € threshold for world final demand is crossed towards the end of the century in FST1 if land-related limits to growth are ignored. However, the application of land-related limits leads to a much slower growth in FST1 (only 140 trillion € of final demand in 2100). Demographic changes in both scenarios are comparable until 2050 and start to diverge afterwards, with SSP1 projecting a 1.1 billion smaller world population by 2100 relative to FST1 (Samir and Lutz, 2017). Total final energy use by 2100 are fairly similar (540 EJ for SSP1, 555 EJ for FST1), with energy per capita being moderately higher in SSP1 (77 GJ per capita) than in FST1 (68 GJ per capita). The electrification rate of the energy system in FST1 reaches 72% by 2100, which lies within the upper range of electrification levels reported for SSPs with low emission trajectories (Kriegler et al., 2018). Finally, while FST1 is associated with a temperature increase of 2.6 °C until the end of the century, SSP1 without additional climate policies results in a 3 °C increase. Thus, the biggest difference between FST1 and SSP1 lies in the much lower rate of economic growth.

Quantified scenarios closest to FST5 are the LED (Low energy demand) scenario (Grubler et al., 2018b) as well as the DLE (Decent living energy) scenario (Millward-Hopkins et al., 2020), due to their focus on sufficiency. FST5, LED and DLE share low total final energy use as well as low energy per capita levels: In 2050, total final energy amounts to 236 EJ in FST5, slightly below the 245 EJ reported for DLE, and above the 149 EJ reported for LED. However, by 2100, total final energy has declined to 114 EJ in FST5, which is considerably lower than the DLE level, as well as the LED level for 2100 (235 EJ). Likewise, final energy per capita ranges between 23 GJ/cap (in 2050) and 29 GJ/cap (in 2100) in FST5, which aligns well with the energy per capita values in LED (26 to 27 GJ per capita) as well as with the per capita values of the ‘Less-advanced-technology’ scenario variant of the DLE simulations (26 GJ/cap). The similarity in results across FST5, LED, and DLE is remarkable, given the top-down nature of the FST5 simulation with MORDRED. Likewise, electrification evolves comparably in FST5 and LED, with the electrification rate reaching 55% in LED and 48% in FST5 by 2100. Last, both LED and FST5 manage to keep global warming below 2 °C.

Despite these similarities, two differences set the FST5 simulation apart from the LED and the DLE scenario. First, while FST5 assumes a strong reduction in the size of the world population, LED and DLE assume that world population does not decline relative to current levels. Second, while the DLE scenario does not include GDP projections, the LED scenario database reports global GDP increasing to 539 trillion USD by 2100. This contrasts sharply with the global final demand value of 55 trillion € at the end of the FST5 simulation period. Thus, FST5 constitutes an alternative sustainability transition that has not yet been explored in the literature.

The evolution of energy per capita and the share of electricity in the energy mix are graphically depicted for FST1 and FST5 in Section 3 of SM1 while the evolution of final demand is depicted in Section 1.1 of SM1.

4. Discussion

4.1. Boundary conditions and scenario plausibility

In order to test to which extent key model dynamics differ when important boundary conditions are altered, we conducted three additional simulations for both FST scenarios. In each additional simulation, certain scenario parameters were modified. Changing key boundary conditions in the simulations serves to identify which scenario assumptions must hold for the outcomes to resemble the FST storylines. These assumptions can then be evaluated in terms of their plausibility.

The first scenario variation (SV1), which is simulated for both FST1 and FST5, explores the effect of higher climate sensitivity and lower GHG reductions on the stability of simulation results. Therefore, we reduced non-combustion-related GHG intensities by 10% instead of 35%, and increased the sensitivity of the climate system to anthropogenic pressure - specifically by increasing the equilibrium climate sensitivity (ECS) by 10% and assuming a higher sensitivity of natural CH₄ emissions and of carbon uptake by biomass from the atmosphere to temperature changes (cf. Kauffhold et al., 2025).

In SV2, also simulated for both FST1 and FST5, the reduction in the land intensity of renewable energy sectors is set to 10% by 2100, i.e. it is assumed that only limited land-use savings can be achieved in renewable energy generation without reducing technological performance (Capellán-Pérez et al., 2017).

SV3, simulated only for FST1, we assess the consequences of lower demand for land, resulting from both reduced land intensities

Table 2
Overview of scenario variations.

Scenario variation	FST1	FST5
SV1	Higher climate sensitivity and less GHG intensity reductions.	
SV2	Lower reductions in the land intensity of renewable energy sectors.	
SV3	Less pressure on land.	
SV4		Greater reduction in labor productivity.

and a lower reliance on biomass-based renewable energy. Compared to the ‘default’ FST1 run, the target land intensity in 2100 is reduced by 50% for all non-energy-related land uses. Additionally, it is assumed that only 5% of non-electric fossil energy is substituted by biomass.

SV4, simulated only for FST5, explores the effects of a greater reduction in labor productivity on scenario outcomes. Accordingly, while global average sectoral labor intensities increase by 10% by 2100 in the original scenario, in this variation they are set to increase by 50% (Table 2).

For each dimension (economic, social, environmental) we select one key variable for the assessment: *output*, because it represents the scale of overall economic activity; *consumption per capita*, because in the model this variable influences life expectancy and serves as an indicator of well-being (at least until a certain consumption level) (Carver and Grimes, 2019; Vollebregt et al., 2024); and *changes in global average temperature*, because they serve as an indicator of anthropogenic pressure on planetary systems.

Simulating the first scenario variation (SV1) we find that more pessimistic assumptions about the climate system’s sensitivity lead to higher levels of global warming, lower output due to climate damages, and reduced per capita consumption (Fig. A.1). The output loss is significantly greater for FST1 than for FST5, and it causes the consumption of the richest class to fall below their initial level by 2100, while the reduction in consumption of the poorest class further hampers poverty elimination. Because even small changes in the magnitude of global warming visibly alter the scenario outcomes, we conclude that the quantifications—especially for FST1—are sensitive to assumptions about the climate system, and that greater climate sensitivity diminishes the scenarios’ long-term stability and renders them less plausible.

In SV2 we observe relatively small deviations from the original scenario outcomes (Fig. A.2e–h); these deviations are reduced in FST1 and nearly eliminated in FST5 during the second half of the simulation. Interestingly, this suggests that technological progress in (land) efficiency of renewables has a comparatively minor influence on the feasibility of the scenarios.

SV3 diverges notably from the original FST1 simulation, with higher outcomes, elevated consumption levels and weak climate mitigation. Despite the substantial improvement in land intensity—which buffers the increasing climate-change-related land damages—growing difficulties in sustaining economic growth become apparent toward the end of the simulation. Since a low level of fossil–bioenergy substitution does not align with the FST1 storyline, and the very high land-intensity improvements required to achieve the same growth rate through biomass-based energy seem difficult to attain (especially given factors that tend to reduce land productivity, such as soil-fertility loss (Cherlet et al., 2018) or forest productivity decline (Mohr et al., 2025)), FST1 appears plausible (i.e. internally consistent and compatible with current knowledge) only as a ‘low-growth’ scenario with average annual growth rates in world output of under 1%.

For FST5, SV4 demonstrates that the assumed increase in labor intensity in the original quantification represents an upper bound: with lower labor productivity levels, the consumption target of FST5 can no longer be met (Fig. A.2h). Thus, assuming a stronger increase in labor intensity - as in SV4 - reduces the feasibility of FST5.

There are other important boundary conditions that we did not vary in the scenario analyses. These include resource availability, the severity of damages, and impacts of biodiversity loss. Estimates of remaining and accessible fossil resources vary widely (Delannoy et al., 2021; Fricko et al., 2017; Laherrère et al., 2022; Turner, 2008). However, using a medium estimate in our simulations, we find that by 2100 only 74% of estimated fossil reserves have been extracted in FST1 and only 45% in FST5. Thus, in sustainability transition scenarios, uncertainty about fossil resources plays a lesser role than in high-emission pathways as adopting more optimistic assumptions regarding future accessibility of fossil resources will not impact the simulation outcomes.

Regarding the accessibility of minerals, comparing the quantities of tin and copper extracted between 2019 and 2100 in FST1 and FST5 with the Ultimately Available Resources (UAR) estimates of Henckens (2021), we find extracted shares of 18% and 5% for tin, and 27% and 8% for copper, respectively. We conclude that mineral resource availability has a relatively minor impact on scenario plausibility - especially for FST5.

The severity of climate damages was not varied in the scenarios, although they certainly influence development pathways (IPCC, 2022). Nevertheless, since damages evolve nonlinearly, we argue that the question of severity is more central for worst-case climate scenarios, while the transition scenarios modeled here exhibit comparatively moderate temperature rises. Hence, neither an increase nor a decrease in the assumed climate damage parameters can be expected to significantly alter simulation outcomes.

Finally, given that MORDRED does not represent potential effects of declining ecosystem services on economic reproduction the scenario outcomes are likely over-optimistic, especially for FST1 which exhibits higher resource and land pressures. At the same time, the model might overestimate land-related limits to growth in the FST1 scenario, partly due to a relatively fixed share of demanded land-intensive products even under emerging land-related limits to growth (see Section 3 SM1 for a discussion of land-related uncertainties).

Overall, it is important to stress that, while the scenario variations and the discussion of uncertainty regarding environmental feedback on economic development serve to contextualize the simulation results, they should not be interpreted as comprehensive sensitivity analyses. Rather, they show that there are certain corridors of parameter values within which the simulation results fit the underlying qualitative scenario storylines. The existence of these value corridors points to the plausibility of FST1 and FST5 storylines, i.e. they underscore their internal consistency and compatibility with current knowledge of the state of the world system.

Comparing FST1 and FST5 with different feasibility considerations developed in the literature (Beath et al., 2025; Brutschin et al., 2021) we find that there are low geophysical feasibility concerns (regarding mineral availability and technical potential of renewable energy sources) for both scenarios, as well as low technological concerns for FST5, while FST1 approaches the medium-concern level for the rate of scale-up of wind and solar technologies during the first decades of the simulated sustainability transition. Additionally, environmental feasibility concerns regarding bioenergy could apply for the last decades of the FST1 simulation. Since neither FST1 nor FST5 rely on carbon dioxide removal technologies, no feasibility thresholds are applicable. At the same time, the electrification rate falls within the lower range of IAM-based scenarios simulations (Beath et al., 2025); hence, feasibility concerns are low for FST1 and FST5 compared to SSP-based transition scenarios. Regarding socio-cultural feasibility, for FST1 the maximum rate of decline in (per capita) energy demand remains below the medium concern threshold defined by Brutschin et al. (2021), whereas for FST5 the high concern threshold is crossed, but only during the first decade of the simulated global transition. Moreover, while institutional feasibility is not properly accounted for in SSP-based scenarios (Brutschin et al., 2021), in the FST scenario framework, institutional feasibility issues are addressed through the qualitative storylines that ensure consistency between the modeled policies and the envisioned global institutional environment.

4.2. Policy relevance and future work

Our simulations are not predictions of the future claiming ‘quantitative certainty’ but data-based imagined futures supporting societal and political decision-making and planning (Juri et al., 2025). As we focus on very broad development at the global level, the policy relevance of our work equally is of global character since regional or local policies can be better informed by concrete case-specific modeling studies. Three key policy-relevant insights can be formulated based on the simulation outcomes of FST1 and FST5.

First, the modeling results clearly indicate that FST5 has better chances to meet both environmental and social sustainability goals, namely a slowdown of environmental degradation, the eradication of poverty and the reduction of social inequality. FST1 does not only fail to achieve climate goals but is also characterized by a constant decrease in the workforce utilization rate, pointing at a risk of emerging unemployment issues due to constantly rising labor productivity, which has already been discussed by O’Neill (2020). Thus, our work provides model-based support for those strands in the ongoing debate on sustainability transitions that promote development paradigms and policies similar to those in FST5, such as degrowth, post-growth or steady-state economics (Kallis et al., 2025; Slameršak et al., 2026).

Second, the simulations show that strong reductions of anthropogenic pressure on planetary systems are possible even in the absence of speculative technologies or technological breakthroughs such as negative emission technologies or nuclear fusion. However, as shown by the FST5 modeling results, successful mitigation depends critically on radical convergence, consumption reduction and a series of changes in production processes such as recycling, product reuse etc. Importantly, in the simulation, policy measures are implemented fast and throughout the world. Thus, our scenario study confirms the urgent need to look for radical solutions beyond national boundaries, which in turn, can only be implemented under a changed world order with new institutional ‘rules of the game’ shaping global economic, political and cultural systems.

Last, the scenario quantification exercise reaffirms the importance of holistic sustainability policies targeting economic, social, cultural, demographic as well as technological developments, since it is the combination of simultaneous changes in different sub-modules that drive simulation outcomes in both FSTs. For example, demographic changes in FST5 significantly reduce the required scale of climate mitigation. Likewise, the rapid decarbonization of the economy in both scenarios increases the relative importance of GHG emissions from other sources than fossil fuel combustion, such as N₂O or SF₆ that should be addressed with specific globally harmonized policies.

At the same time, it is important to stress that the simulated policies unfold in a radically changed institutional and political environment described in the respective storylines. In the context of the current world order and unequal distribution of power, discourses about demography-related climate mitigation entail the risk of being misused to justify neocolonial practices of influencing population policies in the peripheral countries, or to shift the debate from climate justice and economic inequality to population growth in the Global South. However, under a ‘planetary confederation’ as proposed by the FST5 storyline, a combination of policies which is unfeasible under the present institutional constellation - namely voluntary birth rate reductions in combination with strong inter- and intranational reductions in inequality, becomes conceivable, and therefore it is meaningful to explore the policy effects through quantitative modeling.

Arguably, the assumed reduction of world population in FST5 is quite drastic and could be considered an ‘extreme’ case. Thus, future work could explore alternative parametrizations, determining the rates of required changes in consumption, reuse, recycling and substitution of fossil fuels compatible with achieving climate goals with a larger world population.

Likewise, our simulations could be refined in future work by integrating more biophysical feedbacks to address the optimistic bias of the simulations, by improving the representation of land-related processes, and by updating the IO data used (Stadler et al., 2018). Furthermore, in order to allow for a more nuanced modeling of distributional policies, the representation of inequality could be further enhanced, given that the upper and lower tails of income and wealth distribution tend to follow a power law (Safari et al., 2020; Tian and Liu, 2020).

Equally, multiple FST scenarios, including FST1 and FST5, could be introduced into different IAMs. This would enable comparing the performance of different sustainability paradigms in models other than MORDRED but also would serve to explore differences in outcomes stemming from differences in model structures. Since many IAMs use negative emission technologies it can be expected that climate-related results would improve for both FSTs. Additionally, the emission pathways generated by MORDRED could be fed into more sophisticated climate modules to explore possible variations in the level of global warming. Finally, researchers could develop new models that focus on aspects of the FSTs that neither MORDRED nor other existing IAMs can represent, such as ownership structures, decision-making processes in economic production, basic human needs and a need-based allocation of goods (Table 1).

5. Conclusion

In this article, we quantified two global sustainability transition scenarios—FST1: Greener Growth and FST5: Sufficiency Economy—using the MORDRED-IAM. Overall, the scenario outcomes align well with their respective storylines, supporting their plausibility. The simulations complement the qualitative narratives by generating demographic, consumption, production, and emission pathways. Both scenarios achieve substantial decarbonization without relying on speculative technologies; however, only FST5 limits global warming to below 2 °C. Future research could extend this work by quantifying the remaining FSTs, implementing FST1 and FST5 in other IAMs, and developing new models to explore aspects not covered in this analysis, such as shifts in ownership structures or need-based goods allocation.

CRediT authorship contribution statement

Arthur Lauer: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Luis Llases:** Software, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.eist.2026.101157](https://doi.org/10.1016/j.eist.2026.101157).

Appendix

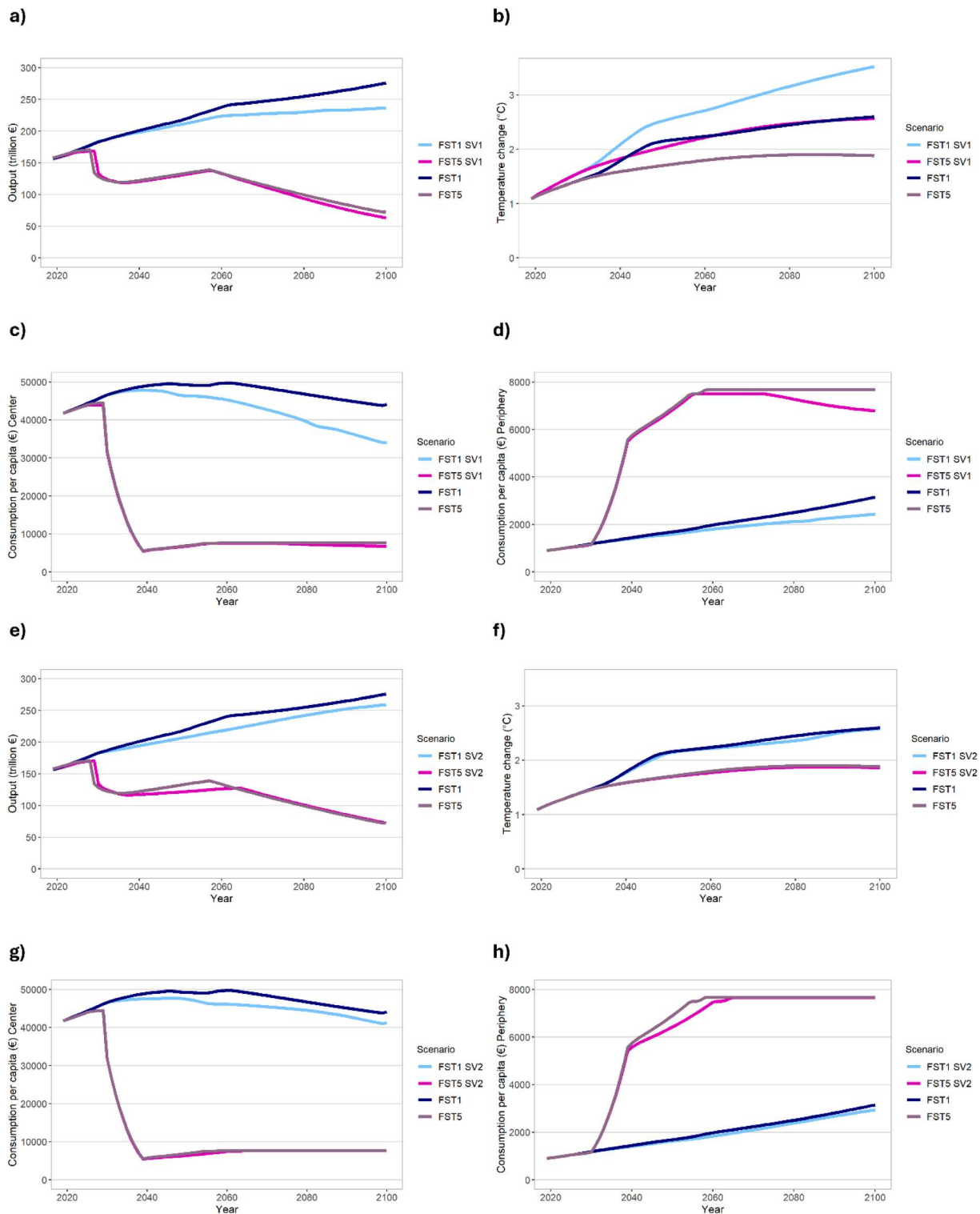


Fig. A.1. Evolution of key output variables in scenarios with changed boundary conditions compared to the baseline scenario results. a & e: total world output; b & f: global average temperature change compared to 1850; c & g: consumption per capita of the richest 20% of the center; d & h: consumption per capita of the poorest 30% of the periphery. (a–d) Baseline scenarios compared to scenarios with higher climate sensitivity and less GHG intensity reductions; (e–h) Baseline scenarios compared to scenarios with lower reductions in the land intensity of renewable energy sectors.

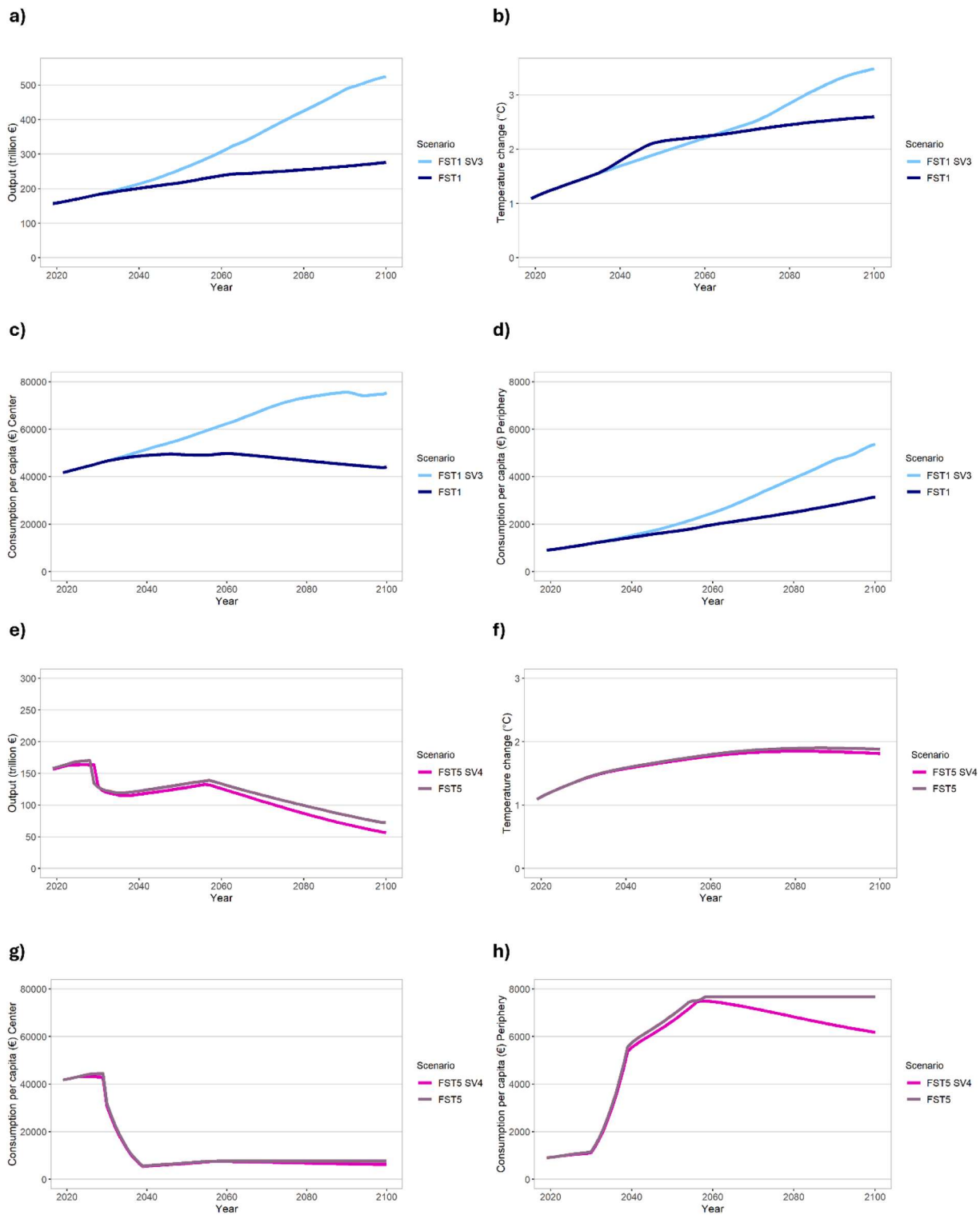


Fig. A.2. Evolution of key output variables in scenarios with changed boundary conditions compared to the baseline scenario results. a & e: total world output; b & f: global average temperature change compared to 1850; c & g: consumption per capita of the richest 20% of the center; d & h: consumption per capita of the poorest 30% of the periphery. (a–d) Baseline compared to a scenario with higher land intensity reductions and lower substitution of fossil energy through bioenergy for FST1. (e–h) Baseline compared to a scenario with a higher labor intensity increase for FST5.

Data availability

Data will be made available on request.

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